General Bite #10: Feed Lines and Basic Antennas

Coax loss, SWR math, characteristic impedance; then dipoles, ground-planes, length calculations, and antenna height effects.

Groups G9A + G9B · 23 pool questions · 5-10 minute presentation · Practice exams at eham.net/exams

Part 1: Feed Lines (G9A)

Characteristic Impedance And SWR

Question	Answer	Q
Factors that determine Z_0 of parallel conductor line	**The distance between conductor centers and the radius of the conductors** (G9A01).	G9A01
Z_0 of "window line"	**450 ohms** (G9A03).	G9A03
Cause of reflected power at the antenna feed point	**A difference between feed line impedance and antenna feed point impedance** (G9A04). Mismatch = reflection.	G9A04
How to prevent standing waves on a feed line	**The antenna feed point impedance must be matched to the characteristic impedance of the feed line** (G9A07).	G9A07

SWR Calculation

SWR for a purely resistive mismatch: $Z_{high} \div Z_{low}$.
G9A09: 50 Ω feed line into 200 Ω resistive load → 200 ÷ 50 = **4:1** (G9A09).
G9A10: 50 Ω feed line into 10 Ω resistive load → 50 ÷ 10 = **5:1** (G9A10). Note: SWR is always written as the larger number first, so it's 5:1, not 1:5.

Loss And Frequency

Question	Answer	Q
How coax attenuation changes with frequency	**Attenuation increases** with frequency (G9A05). Higher frequency = more skin-effect and dielectric loss.	G9A05
Units of RF feed line loss	**Decibels per 100 feet** (G9A06).	G9A06
Relationship between high SWR and line loss	**High SWR increases loss in a lossy transmission line** (G9A02). Reflected power makes extra trips through the lossy line.	G9A02
What does a tuner do to SWR on the feed line?	**If line SWR is 5:1 and the tuner shows 1:1 to the transmitter, the line SWR is still 5:1** (G9A08). The tuner masks the mismatch from the transmitter but does not reduce the actual line SWR.	G9A08
Effect of line loss on SWR measured at the input	**Higher loss reduces SWR measured at the input to the line** (G9A11). Loss attenuates both forward and reflected; SWR meter reads better than reality.	G9A11

Part 2: Basic Dipole And Monopole Antennas (G9B)

Random Wire And Ground Plane

Question	Answer	Q
Characteristic of random-wire antenna fed directly	**Station equipment may carry significant RF current** (G9B01). No counterpoise → RF returns through the rig.	G9B01
How to adjust ground-plane vertical to ≈ 50 Ω	**Slope the radials downward** (G9B02). Drooping radials raise feed-point impedance from 36 Ω to 50 Ω.	G9B02
Radiation pattern of a 1/4-wave ground-plane vertical	**Omnidirectional in azimuth** (G9B03).	G9B03

Dipole Patterns And Heights

Question	Answer	Q
Pattern of a free-space dipole in the plane containing the conductor	**A figure-eight at right angles to the antenna** (G9B04). Lobes broadside, nulls off the ends.	G9B04
Azimuthal pattern of horizontal dipole at high angles, when antenna is below 1/2 λ height	**Almost omnidirectional** (G9B05). Low-height dipole's high-angle radiation has no clear lobes.	G9B05
Radial placement for ground-mounted vertical	**On the surface or buried a few inches below the ground** (G9B06).	G9B06
Feed-point impedance of horizontal 1/2-wave dipole as antenna is lowered to 1/10 λ height	**It steadily decreases** (G9B07). Ground proximity drops the Z below the free-space 73 Ω.	G9B07

Feed-point impedance of 1/2-wave dipole as feed is moved toward the ends	**It steadily increases** (G9B08). Voltage maximum at the ends, current minimum.	G9B08
Advantage of horizontal vs vertical HF polarization	**Lower ground losses** (G9B09).	G9B09

Dipole Length Math

Half-wave dipole length (ft) ≈ 468 / frequency in MHz.
Quarter-wave monopole length (ft) ≈ 234 / frequency in MHz.
G9B10: 1/2 λ at 14.250 MHz → 468 / 14.25 ≈ **33 ft** (G9B10).
G9B11: 1/2 λ at 3.550 MHz → 468 / 3.55 ≈ **132 ft** (G9B11).
G9B12: 1/4 λ at 28.5 MHz → 234 / 28.5 ≈ **8 ft** (G9B12).

Mental Cheat Sheet

- Parallel line Z_0 is determined by conductor radius and spacing (ratio, not sum).
- Window line Z_0 = **450 Ω**.
- SWR resistive = Z_{high}/Z_{low}. 50→200 = **4:1**; 50→10 = **5:1**.
- Coax loss **increases** with frequency.
- Loss expressed in **dB per 100 ft**.
- High SWR → **more loss** in a lossy line.
- Tuner gives TX 1:1 but line SWR unchanged.
- Line loss **reduces** SWR seen at input.
- Random wire fed directly → RF on equipment.
- Ground-plane to 50 Ω: **droop the radials**.
- 1/4 λ vertical = omnidirectional in azimuth.
- Dipole pattern (in conductor plane) = **figure-8 broadside**.
- Low dipole high-angle pattern: almost omni.
- Ground radials: **on surface or just-buried**.
- Low dipole → Z **decreases**.
- Feed off-center → Z **increases** toward ends.
- Horizontal vs vertical HF = lower **ground losses**.
- 1/2 λ ft = **468 / f(MHz)**.
- 1/4 λ ft = **234 / f(MHz)**.

Common Traps

- Parallel-line Z_0 depends on **spacing and radius**, not length or frequency.
- SWR 50 → 200 Ω is **4:1**, not 2:1.
- SWR 50 → 10 Ω is **5:1**, not 1:5 (always larger/smaller).
- Coax loss **increases** with frequency, not "independent of frequency."
- The transmitter sees the tuner's 1:1, but **the line itself still has 5:1 SWR**.
- Line loss **reduces** the SWR you read at the input; the antenna mismatch is actually higher.
- To raise ground-plane Z to 50 Ω, **droop radials downward**, not upward.
- Dipole pattern (in conductor plane) is **figure-8 broadside**, not off the ends.
- Low dipole impedance **decreases** with height drop, not increases.
- Off-center feed **raises** Z; never **decreases**.
- 1/2 λ dipole length uses 468/f, **not** 234/f (that's quarter-wave).

Takeaway: Feed-line impedance comes from conductor geometry; window line is 450 Ω. SWR (resistive) = Z_{high}/Z_{low}. Tuners hide mismatch from the transmitter but do not fix line SWR. Coax loss goes up with frequency and with SWR. Dipoles radiate broadside; lowering them drops feed impedance. Length math: 468/f for 1/2 λ, 234/f for 1/4 λ (frequency in MHz, answer in feet).

Bite 10 Question Bank (G9A + G9B)

G9A01. Which of the following factors determine the characteristic impedance of a parallel conductor feed line?
A. The distance between the centers of the conductors and the radius of the conductors.
B. The distance between the centers of the conductors and the length of the line.
C. The radius of the conductors and the frequency of the signal.
D. The frequency of the signal and the length of the line.

G9A02. What is the relationship between high standing wave ratio (SWR) and transmission line loss?
A. There is no relationship between transmission line loss and SWR.
B. High SWR increases loss in a lossy transmission line.
C. High SWR makes it difficult to measure transmission line loss.
D. High SWR reduces the relative effect of transmission line loss.

G9A03. What is the nominal characteristic impedance of "window line" transmission line?
A. 50 ohms.
B. 75 ohms.
C. 100 ohms.
D. 450 ohms.

G9A04. What causes reflected power at an antenna's feed point?
A. Operating an antenna at its resonant frequency.
B. Using more transmitter power than the antenna can handle.
C. A difference between feed line impedance and antenna feed point impedance.
D. Feeding the antenna with unbalanced feed line.

G9A05. How does the attenuation of coaxial cable change with increasing frequency?
A. Attenuation is independent of frequency.
B. Attenuation increases.
C. Attenuation decreases.
D. Attenuation follows Marconi's Law of Attenuation.

G9A06. In what units is RF feed line loss usually expressed?
A. Ohms per 1,000 feet.
B. Decibels per 1,000 feet.
C. Ohms per 100 feet.
D. Decibels per 100 feet.

G9A07. What must be done to prevent standing waves on a feed line connected to an antenna?
A. The antenna feed point must be at DC ground potential.
B. The feed line must be an odd number of electrical quarter wavelengths long.
C. The feed line must be an even number of physical half wavelengths long.
D. The antenna feed point impedance must be matched to the characteristic impedance of the feed line.

G9A08. If the SWR on an antenna feed line is 5:1, and a matching network at the transmitter end of the feed line is adjusted to present a 1:1 SWR to the transmitter, what is the resulting SWR on the feed line?
A. 1:1.
B. 5:1.
C. Between 1:1 and 5:1 depending on the characteristic impedance of the line.
D. Between 1:1 and 5:1 depending on the reflected power at the transmitter.

G9A09. What standing wave ratio results from connecting a 50-ohm feed line to a 200-ohm resistive load?
A. 4:1.
B. 1:4.
C. 2:1.
D. 1:2.

G9A10. What standing wave ratio results from connecting a 50-ohm feed line to a 10-ohm resistive load?
A. 2:1.
B. 1:2.
C. 1:5.
D. 5:1.

G9A11. What is the effect of transmission line loss on SWR measured at the input to the line?
A. Higher loss reduces SWR measured at the input to the line.
B. Higher loss increases SWR measured at the input to the line.
C. Higher loss increases the accuracy of SWR measured at the input to the line.
D. Transmission line loss does not affect the SWR measurement.

G9B01. What is a characteristic of a random-wire HF antenna connected directly to the transmitter?
A. It must be longer than 1 wavelength.
B. Station equipment may carry significant RF current.
C. It produces only vertically polarized radiation.
D. It is more effective on the lower HF bands than on the higher bands.

G9B02. Which of the following is a common way to adjust the feed point impedance of an elevated quarter-wave ground-plane vertical antenna to be approximately 50 ohms?
A. Slope the radials upward.
B. Slope the radials downward.
C. Lengthen the radials beyond one wavelength.
D. Coil the radials.

G9B03. Which of the following best describes the radiation pattern of a quarter-wave ground-plane vertical antenna?
A. Bi-directional in azimuth.
B. Isotropic.
C. Hemispherical.
D. Omnidirectional in azimuth.

G9B04. What is the radiation pattern of a dipole antenna in free space in a plane containing the conductor?
A. It is a figure-eight at right angles to the antenna.
B. It is a figure-eight off both ends of the antenna.
C. It is a circle (equal radiation in all directions).
D. It has a pair of lobes on one side of the antenna and a single lobe on the other side.

G9B05. How does antenna height affect the azimuthal radiation pattern of a horizontal dipole HF antenna at elevation angles higher than about 45 degrees?
A. If the antenna is too high, the pattern becomes unpredictable.
B. Antenna height has no effect on the pattern.
C. If the antenna is less than 1/2 wavelength high, the azimuthal pattern is almost omnidirectional.
D. If the antenna is less than 1/2 wavelength high, radiation off the ends of the wire is eliminated.

G9B06. Where should the radial wires of a ground-mounted vertical antenna system be placed?
A. As high as possible above the ground.
B. Parallel to the antenna element.
C. On the surface or buried a few inches below the ground.
D. At the center of the antenna.

G9B07. How does the feed point impedance of a horizontal 1/2 wave dipole antenna change as the antenna height is reduced to 1/10 wavelength above ground?
A. It steadily increases.
B. It steadily decreases.
C. It peaks at about 1/8 wavelength above ground.
D. It is unaffected by the height above ground.

G9B08. How does the feed point impedance of a 1/2 wave dipole change as the feed point is moved from the center toward the ends?
A. It steadily increases.
B. It steadily decreases.
C. It peaks at about 1/8 wavelength from the end.
D. It is unaffected by the location of the feed point.

G9B09. Which of the following is an advantage of using a horizontally polarized as compared to a vertically polarized HF antenna?
A. Lower ground losses.
B. Lower feed point impedance.
C. Shorter radials.
D. Lower radiation resistance.

G9B10. What is the approximate length for a 1/2 wave dipole antenna cut for 14.250 MHz?
A. 8 feet.
B. 16 feet.
C. 24 feet.
D. 33 feet.

G9B11. What is the approximate length for a 1/2 wave dipole antenna cut for 3.550 MHz?
A. 42 feet.
B. 84 feet.
C. 132 feet.
D. 263 feet.

G9B12. What is the approximate length for a 1/4 wave monopole antenna cut for 28.5 MHz?
A. 8 feet.
B. 11 feet.
C. 16 feet.
D. 21 feet.

Answer Key

G9A01: A	G9A02: B	G9A03: D	G9A04: C	G9A05: B
G9A06: D	G9A07: D	G9A08: B	G9A09: A	G9A10: D
G9A11: A	G9B01: B	G9B02: B	G9B03: D	G9B04: A
G9B05: C	G9B06: C	G9B07: B	G9B08: A	G9B09: A
G9B10: D	G9B11: C	G9B12: A		